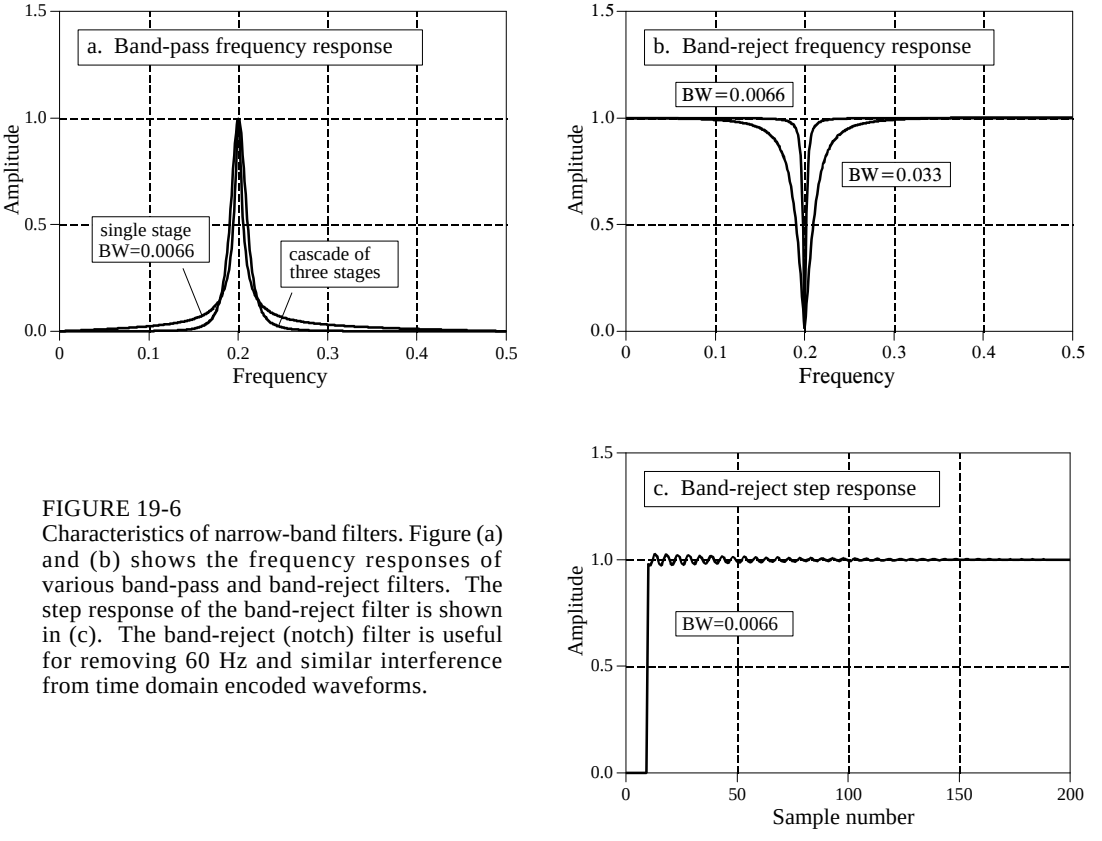

FIGURE 19-6
Characteristics of narrow-band filters. Figure (a) and (b) shows the frequency responses of various band-pass and band-reject filters. The step response of the band-reject filter is shown in (c). The band-reject (notch) filter is useful for removing 60 Hz and similar interference from time domain encoded waveforms.

Two parameters must be selected before using these equations: *f*, the center frequency, and *BW*, the bandwidth (measured at an amplitude of 0.707). Both of these are expressed as a fraction of the sampling frequency, and therefore must be between 0 and 0.5. From these two specified values, calculate the intermediate variables: *R* and *K*, and then the recursion coefficients.

As shown in (a), the band-pass filter has relatively large *tails* extending from the main peak. This can be improved by cascading several stages. Since the design equations are quite long, it is simpler to implement this cascade by filtering the signal several times, rather than trying to find the coefficients needed for a single filter.

Figure (b) shows examples of the band-reject filter. The narrowest bandwidth that can be obtain with single precision is about 0.0003 of the sampling frequency. When pushed beyond this limit, the attenuation of the notch will degrade. Figure (c) shows the step response of the band-reject filter. There is noticeable overshoot and ringing, but its amplitude is quite small. This allows the filter to remove narrowband interference (60 Hz and the like) with only a minor distortion to the time domain waveform.